Kochen–Specker uncolorability in cyclotomic fields requires exactly $6 \mid n$

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(Dated: February 18, 2026)

We prove that the ray set generated by the nth roots of unity in $\mathbb{C}^3$ is Kochen–Specker uncolorable if and only if $6 \mid n$. Sufficiency follows from the embedding of Cabello's Eisenstein 33-vector KS set. Necessity is proved by algebraic analysis of vanishing sums among roots of unity: when n is odd and $3 \mid n$, a projective collapse argument shows every triad is isolated (no two share a ray), so the coloring problem decomposes; when $3 \nmid n$ and $2 \mid n$, the orthogonality graph on each coordinate plane is a perfect matching, admitting an explicit coloring. The factorization $6 = 2 \times 3$ is consistent with the result of Cortez, Morales, and Reyes that $\mathbb{Z}[1/6]$ is the minimal ring supporting KS contextuality in dimension 3: the arithmetic conditions $2 \mid n$ (two-term cancellation) and $3 \mid n$ (three-term phase cancellation) are independently necessary and jointly sufficient.

INTRODUCTION

The Kochen–Specker (KS) theorem [1] establishes that quantum measurements in dimension $d \geq 3$ cannot be explained by noncontextual hidden-variable theories, by exhibiting a finite set of rays admitting no consistent $\{0, 1\}$-valued coloring. A natural source of KS constructions is the cyclotomic family: for each positive integer n, the alphabet $\{0\} \cup \{\zeta^k : k = 0, \dots, n-1\}$ with $\zeta = e^{2\pi i/n}$ generates a ray set S_n in $\mathbb{C}^3$ via all projectively distinct nonzero vectors with coordinates drawn from this alphabet.

Cabello [2] showed that S_6 contains a 33-vector KS set built from the Eisenstein integers $\mathbb{Z}[\omega]$ ($\omega = e^{2\pi i/3}$), which he identified as the "simplest" KS set by the criterion of fewest bases. Cortez, Morales, and Reyes [3] proved that $\mathbb{Z}[1/6]$ is the minimal ring extension of $\mathbb{Z}$ exhibiting KS contextuality in dimension 3—that is, the primes 2 and 3 are both necessary.

These results suggest a sharp characterization: KS-uncolorability in the cyclotomic family should require exactly $6 \mid n$. We prove this.

Theorem 1. *The ray set S_n generated by the nth roots of unity in $\mathbb{C}^3$ is KS-uncolorable if and only if $6 \mid n$.*

PRELIMINARIES

A *ray* in $\mathbb{C}^3$ is a one-dimensional subspace, represented by a nonzero vector up to scalar multiples. Two rays v, w are *orthogonal* if $\langle v|w\rangle = \sum_k \bar{v}_k w_k = 0$. An orthogonal *triad* is a set of three mutually orthogonal rays (an orthonormal basis up to normalization). A *KS coloring* assigns each ray a value in $\{0, 1\}$ such that (I) orthogonal rays cannot both receive 1, and (II) every triad contains exactly one 1. A ray set is *KS-uncolorable* if no such coloring exists.

For $\zeta = e^{2\pi i/n}$, the ray set S_n consists of all projectively distinct (modulo $\mathbb{C}^\times$) nonzero vectors with coordinates in $\{0\} \cup \{\zeta^j : 0 \leq j < n\}$. We classify rays by *zero pattern*: *axis rays* have two zero coordinates (3 rays); *one-zero rays* have exactly one zero ($3n$ rays: normalizing the first nonzero coordinate to 1 leaves one free phase, and there are 3 choices of zero position); *all-nonzero rays* have no zeros (n^2 rays: normalizing the first coordinate to 1 leaves two free phases).

A key structural observation: in $\mathbb{C}^3$, the inner product $\langle v|w\rangle = \sum_k \bar{v}_k w_k$ has at most three nonzero terms. Each nonzero term is $\overline{\zeta^a}\zeta^b = \zeta^{b-a}$, an nth root of unity. Therefore only 1-, 2-, and 3-term vanishing sums of nth roots arise as orthogonality conditions in S_n.

PROOF OF THEOREM 1

Sufficiency

If $6 \mid n$, then $\zeta^{n/6}$ generates the 6th roots of unity as a subgroup of $\langle\zeta\rangle$, so the alphabet for $n = 6$ embeds into that for general n. Since S_6 contains Cabello's 33-vector KS set [2] and KS-uncolorability is monotone (any superset of an uncolorable set is uncolorable), S_n is KS-uncolorable. □

Necessity: vanishing-sum analysis

We show that if $6 \nmid n$, then S_n admits a valid KS coloring. The proof rests on two lemmas characterizing when vanishing sums of roots of unity exist.

Lemma 1 (Three-term vanishing sums)**.** *If $u, v, w \in \langle\zeta\rangle$ and $u+v+w = 0$, then $3 \mid n$ and $\{u, v, w\} = \{\lambda, \lambda\omega, \lambda\omega^2\}$ for some $\lambda \in \langle\zeta\rangle$, where $\omega = \zeta^{n/3}$. Conversely, $3 \mid n$ suffices.*

Proof. Write $u = \zeta^a$, $v = \zeta^b$, $w = \zeta^c$. If $a = b$, then $\zeta^a(2 + \zeta^{c-a}) = 0$ forces $|\zeta^{c-a}| = 2 \neq 1$; similarly for other coincidences, so a, b, c are distinct. Dividing by ζ^a: $1 + \zeta^p + \zeta^q = 0$ with $p = b-a$, $q = c-a$

both nonzero. Three unit-modulus numbers summing to zero form an equilateral triangle: $|-1-\zeta^p| = 1$ gives $\cos(2\pi p/n) = -1/2$, hence $p/n \in \{1/3, 2/3\}$, requiring $3 \mid n$. Then $\{1, \zeta^p, \zeta^q\} = \{1, \omega, \omega^2\}$, confirming the classification $\{u, v, w\} = \{\lambda, \lambda\omega, \lambda\omega^2\}$ with $\lambda = \zeta^a$. $\square$

Lemma 2 (Two-term vanishing sums)**.** *The equation $\zeta^a + \zeta^b = 0$ has a solution if and only if $2 \mid n$.*

Proof. $\zeta^a + \zeta^b = 0$ iff $\zeta^{b-a} = -1$ iff $-1 \in \langle\zeta\rangle$ iff $2 \mid n$. $\square$

These lemmas, combined with the zero-pattern classification, determine all orthogonalities in S_n. For each pair of rays, the inner product $\langle v|w\rangle$ is a sum of t roots of unity, where t equals the number of coordinates in which both rays are nonzero:

Ray pair	t	Orth. condition
all-nz vs. all-nz	3	$3 \mid n$
all-nz vs. one-zero	2	$2 \mid n$
one-zero vs. one-zero (same)	2	$2 \mid n$
one-zero vs. one-zero (diff.)	1	never
any vs. axis	0–1	$v_k = 0$

The "diff." case has $t = 1$ because the two zero coordinates are distinct, so two of three inner-product terms vanish; the surviving term has modulus 1 and cannot vanish. The "any vs. axis" case is a structural zero-pattern constraint, independent of n; the three axis rays form a triad for all n.

We now treat the three cases where $6 \nmid n$.

Case 1: $3 \nmid n$, $2 \nmid n$

Any triad requires three mutually orthogonal rays spanning $\mathbb{C}^3$. A non-axis triad must contain at least one ray with fewer than two zero coordinates, hence at least one orthogonal pair whose inner product is a 2-term or 3-term sum of roots. Since $3 \nmid n$ forbids 3-term vanishing sums and $2 \nmid n$ forbids 2-term vanishing sums, no such pair can be orthogonal. The sole triad is the axis triad $\{e_1, e_2, e_3\}$, trivially colorable. $\square$

Case 2: $3 \nmid n$, $2 \mid n$

By Lemma 1, no all-nonzero orthogonal pairs exist, so no triad can contain two all-nonzero rays.

Lemma 3 (No all-nonzero ray in any triad when $3 \nmid n$)**.** *If $3 \nmid n$, every triad in S_n contains an axis ray.*

Proof. A triad not containing an axis ray consists of rays each having at most one zero coordinate. Since no two all-nonzero rays are orthogonal ($3 \nmid n$), at most one ray is all-nonzero; the other two are one-zero. Two one-zero rays from *different* planes are never orthogonal (Lemma 5), so both one-zero rays r_1, r_2 share the same zero coordinate k. But then r_1 and r_2 span a subspace of $\{x \in \mathbb{C}^3 : x_k = 0\}$. Since $r_1 \perp r_2$, they form an orthogonal basis for this 2-dimensional subspace, and any vector orthogonal to both must lie in its orthogonal complement: the k-axis. The third ray v is therefore proportional to e_k, contradicting the assumption that v is all-nonzero. A fortiori, three one-zero rays cannot form a triad (the same plane argument applies to any two, and different planes are excluded by Lemma 5). $\square$

Moreover, axis rays are never orthogonal to all-nonzero rays ($e_k \perp v$ requires $v_k = 0$). Since every triad contains an axis ray, no all-nonzero ray appears in any triad. Every triad has the form $\{v_1, v_2, e_k\}$ where v_1, v_2 are one-zero rays in the kth coordinate plane and e_k is the complementary axis ray.

Lemma 4 (Perfect matching)**.** *For even n, the orthogonality graph on one-zero rays in each coordinate plane is a perfect matching.*

Proof. The one-zero rays in the $z = 0$ plane are projectively $(1, \zeta^p, 0)$ for $p \in \{0, \ldots, n-1\}$. Two such rays p, q are orthogonal iff $1 + \zeta^{q-p} = 0$, i.e., $q - p \equiv n/2 \pmod{n}$. Each ray has exactly one partner, giving $n/2$ disjoint edges. $\square$

Lemma 5 (Plane independence)**.** *One-zero rays from different coordinate planes are never orthogonal.*

Proof. If v has $v_i = 0$ and w has $w_j = 0$ with $i \neq j$, then $\langle v|w\rangle = \sum_k \bar{v}_k w_k$ has exactly one nonzero term (at the coordinate $k \neq i, j$), which is a product of roots of unity, hence has modulus 1. For example, $\langle(\zeta^a, \zeta^b, 0)|(\zeta^c, 0, \zeta^d)\rangle = \overline{\zeta^a}\zeta^c + 0 + 0 = \zeta^{c-a} \neq 0$. $\square$

We construct an explicit coloring. Assign $e_1 \mapsto 1$, $e_2, e_3 \mapsto 0$. All one-zero rays in the x=0 plane receive 0 (their triads with e_1 already have one 1). In each of the y=0 and z=0 planes, the matched pairs (Lemma 4) are independently 2-colored: one 1, one 0 per pair. All-nonzero rays receive 0.

Verification of (II): each triad $\{v_1, v_2, e_k\}$ has exactly one 1. For $k = 1$: $e_1 = 1$, both partners 0. For $k = 2, 3$: axis is 0, exactly one of the matched pair is 1.

Verification of (I): we check that no two rays assigned 1 are orthogonal. The ray e_1 is orthogonal only to rays with $v_1 = 0$ (i.e., x=0 plane rays), which all received 0. Rays assigned 1 in the y=0 plane are orthogonal only to their matched partner (assigned 0), to e_2 (assigned 0), and to no ray in the z=0 plane (Lemma 5). The same holds symmetrically for z=0. All-nonzero rays received 0; since only constraint (I) applies to them (they appear in no triad by the above), and 0-valued rays cannot violate (I), the coloring is valid. $\square$

Case 3: $3 \mid n$**,** $2 \nmid n$

This is the principal case. Let $\omega = e^{2\pi i/3} = \zeta^{n/3}$, a primitive cube root of unity. Three-term cancellations create triads among all-nonzero rays, but we show these triads are completely isolated. First, since $2 \nmid n$, no 2-term vanishing sums exist, so no all-nonzero ray is orthogonal to any one-zero ray, and no one-zero ray is orthogonal to another one-zero ray. The all-nonzero and zero-bearing sectors are therefore completely decoupled.

Lemma 6 (Projective collapse)**.** *For any all-nonzero ray v in S_n with $3 \mid n$, there are exactly two projectively distinct all-nonzero rays orthogonal to v.*

Proof. Normalize $v = (1, \zeta^p, \zeta^q)$. For all-nonzero $w = (1, \zeta^r, \zeta^s)$,

$$\langle v|w\rangle = 1 + \zeta^{r-p} + \zeta^{s-q}.$$

By Lemma 1, vanishing requires the three terms to form $\{\lambda, \lambda\omega, \lambda\omega^2\}$. Since one term is 1, we have $\lambda \in \{1, \omega^2, \omega\}$. Each choice determines (r, s) uniquely mod n, giving two projective solutions:

$$w_+ = (1,\ \omega\zeta^p,\ \omega^2\zeta^q),$$
$$w_- = (1,\ \omega^2\zeta^p,\ \omega\zeta^q).$$

(The three λ-values yield only w_+ and w_- up to projective equivalence.) Since $\omega \neq 1$: $w_+ \not\sim v$, $w_- \not\sim v$, and $w_+ \not\sim w_-$. □

Lemma 7 (Triad uniqueness)**.** *$\{v, w_+, w_-\}$ is a triad, it is the unique triad containing v among all-nonzero rays, and $T_+ : v \mapsto w_+$ generates 3-cycles.*

Proof. Orthogonality: $\langle w_+|w_-\rangle = 1+\omega^2\cdot\omega+\omega\cdot\omega^2 = 1+\omega+\omega^2 = 0$ (using $\bar{\omega} = \omega^2$). Uniqueness: v has exactly two all-nonzero orthogonal neighbors (Lemma 6). The map T_+ acts as $(1, \zeta^p, \zeta^q) \mapsto (1, \omega\zeta^p, \omega^2\zeta^q)$. Applying twice: $T_+^2(v) = (1, \omega^2\zeta^p, \omega\zeta^q) = w_-$. Applying thrice: $T_+^3(v) = (1, \zeta^p, \zeta^q) = v$. So each orbit has size exactly 3. □

Lemma 8 (One-zero isolation)**.** *For odd n, no one-zero ray participates in any triad.*

Proof. Consider a triad containing a one-zero ray v with $v_k = 0$. The triad must span $\mathbb{C}^3$, so at least one partner w must have $w_k \neq 0$. Since v has two nonzero coordinates (at positions $\neq k$) and w has $w_k \neq 0$, the inner product $\langle v|w\rangle$ includes a term $\bar{v}_j w_j$ for each position j where both are nonzero. At most one of w's coordinates (other than k) can be zero, so $\langle v|w\rangle$ is a sum of at least 2 roots of unity. A 2-term vanishing sum requires $2 \mid n$ (Lemma 2); a 3-term sum requires $3 \mid n$ (Lemma 1). For odd n, neither condition holds, so $v \not\perp w$: no triad can include a one-zero ray. □

By Lemma 6, the orthogonality graph on all-nonzero rays is 2-regular. By Lemma 7, the map T_+ generates orbits of size exactly 3, each forming a triangle $\{v, w_+, w_-\}$ with all three edges present. Since each vertex has degree 2 and both neighbors lie in its own triangle, there are no edges between triangles. The n^2 all-nonzero rays thus partition into $n^2/3$ isolated triads, plus the axis triad: $n^2/3 + 1$ triads total.

Each triad is isolated: no two all-nonzero triads share a ray (Lemma 7), no one-zero ray participates in any triad (Lemma 8), and no orthogonality edges connect different triads. For the last point: by Lemma 6, each all-nonzero ray has exactly two orthogonal neighbors, both all-nonzero and both in its own triad; and since $2 \nmid n$, no all-nonzero ray is orthogonal to any one-zero ray (2-term sum, requires -1). Thus both constraints (I) and (II) decompose: assign one 1 per triad independently, and assign 0 to all one-zero rays (which are unconstrained). Every triad receives exactly one 1, and no two orthogonal rays both receive 1. □

DISCUSSION

The factorization $6 = 2\times3$ in Theorem 1 reflects two independent algebraic mechanisms, each necessary for KS-uncolorability:

Phase cancellation ($3 \mid n$): the identity $1 + \omega + \omega^2 = 0$ creates orthogonal triads among all-nonzero rays. Without it, no all-nonzero pair is orthogonal, and triads are restricted to zero-coordinate configurations.

Interlocking ($2 \mid n$): the element $-1 \in \langle\zeta\rangle$ creates two-term cancellations that link triads together. Without it, the projective collapse of Lemma 6 ensures every triad is isolated, making the coloring problem decomposable.

This is consistent with the ring-theoretic result of Cortez, Morales, and Reyes [3], who proved that $\mathbb{Z}[1/6]$ is the minimal ring supporting KS contextuality in dimension 3. Their result concerns realizability of projection matrices $P_v = vv^\dagger/\|v\|^2$ over subrings of $\mathbb{Q}$; ours concerns a specific family of coordinate alphabets. The two perspectives are complementary: their $N(S) = \mathrm{lcm}\{\|v\|^2\}$ invariant equals 6 for the Eisenstein 33-set, and Theorem 1 shows that the arithmetic conditions $2 \mid n$ and $3 \mid n$ are independently necessary in the cyclotomic setting as well.

More broadly, the proof technique—classifying orthogonalities by the length of the underlying vanishing sum, then analyzing the resulting constraint graph—may extend to other algebraic families of KS constructions. The projective collapse phenomenon (Lemma 6), where group-theoretic symmetry forces orthogonal neighbors to coalesce, suggests a representation-theoretic approach to KS colorability that goes beyond the combinatorial methods currently dominant in the field.

The author acknowledges the use of Claude (Anthropic) for computational assistance. The proof is self-contained; independent numerical cross-checks using PySAT (Glucose4) verified the coloring constructions for all $n \leq 100$. Code is available at `https://github.com/michaelkernaghan/contextuality`.

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